Unit-1

RANKA
DATE / /
PAGE

- **Biohazards** → Intelligent boys always get very high.
  → biological substances that pose a threat to health of living organism.

1. Infectious Agents
   → disease in human, animal or plant
   → cause to HIV/AIDS, tuberculosis
   e.g Bacteria : Staphylococcus, E.coli
   Viruses : HIV, cold, covid19
   Fungi : ~~Aspergillus~~ (causing infection)
   Parasites : Plasmodium (malaria)

2. Biological Toxin
   → produce toxins by plants, animals
   → poisoning, neurological damage, death
   → contaminate food or water supplies
   e.g. Botulinum toxin : cause botulism
   Ricin :
   Tetrodotoxin : found in puffer fish, death paralysis

3. Allergens
   → allergic reaction
   → breathing, drop in BP, skin, vomit, diarrhea
   Dizziness or fainting
   e.g Pollen : from trees, grasses
   Mold spores : asthma

4. Genetically modified Organisms (GMOs)
   → genes altered using genetic Eng.
   → improve food security
   → can Allergic reactions
   e.g Bt corn & Golden Rice

5. Waste Products
   → Biological waste from farming or industrial processes
   e.g. Manure, Animal remains
   → Medical waste
   e.g. Syringes, ~~Surgical~~ Bandages
   → pathogens, infectious, implanted disposals

6. Vectors
   → Animal to human
   e.g Mosquitoes : carry malaria, dengue
   Rats : bacterium
   Ticks

7. Human & Animal Bodily Fluids
   → Blood, urine, saliva & other fluids with infection
   e.g Blood : HIV, hepatitis B
   Saliva :
   Semen : STI

8. Recombinant DNA Technology
   → DNA molecules that are artificially combined
   e.g Gene therapy drugs

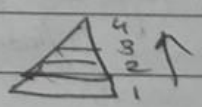
- **BSLs**
  → safety protocols to protect lab, personnel, environment, product
  based on:
  1. Risks related to containment
  2. Severity of infection
  3. Transmissibility
  4. Nature of work conducted within the lab
  5. Origin of microbe
  6. Agent in question
  7. Route of exposure

(1) Biosafety Level 1 (BSL-1)
   → minimal threat
   containment practice: hand wash, use of gloves
   Risk Level : Minimal, agents unlikely to cause disease in healthy individuals.
   e.g E.coli (Non pathogenic)

(2) Biosafety level 2 (BSL-2)
   → moderate risk, have available treatment
   containment Practice: Access to labs restricted, work in BSCs
   → Inhalation, mucous infection
   Eg. Salmonella, HIV

(3) BSL-3
   → High risk but treatment
   containment Practice: BSCs, controlled access & monitored
   e.g SARS-COV-2

(4) BSL-4
   → Extreme risk, dangerous, no treatment
   Containment Practice: Full Isolation, body suit
   e.g Ebola virus.



- **BSCs**
  → specialized ventilation device
  → protect personnel, product & surrounding

(1) Class I
   → protect personnel & environment not product
   → Air is pulled through openings & vents
   → HEPA filter up → recycle
   → Ducted & unducted
   use : suitable for moderate risk (BSL 1, 2).
   Salmonella

Unit - 2

RANKA
DATE / /
PAGE

Cartagena Protocol

→ International treaty aims to ensure safe handling, transport & use of living modified organisms (LMOs) with biotech
→ Adopted on Jan 29, 2000, force on Sep 11, 2003
→ Part of Convention on Biological Diversity (CBD) focuses on managing movement of LMO b/w countries

Objectives:
→ protects biological diversity from adverse effects of LMOs
→ Promote strategies at global, regional & national levels to ensure safe handling
→ Informed decision-making regarding the import of LMOs
→ Rules & procedure.
→ AIA (Advance Informed agreement) b/w countries.
→ notification, Acknowledgement, decision procedure, Review decision.
→ BCH (Biosafety clearing House) serves as an information-sharing mechanism

Focal Areas

① Biosafety systems
→ Establish & further develop effective biosafety systems.
→ Ensure coordination & support
→ Improve risk assessment & risk management
→ protocols for handling, transport, packaging & identification
→ consider socio-economic factors.

② Capacity Building
→ Enhance the ability of countries to implement biosafety measures.
→ Implement national biosafety frameworks.
→ Improve risk assessment & risk management
→ education
→ Information sharing promote

③ Compliance & Review
→ ensure adherence to protocol's provisions
→ mechanism & rules compliance
→ periodic reviews to evaluate effectiveness.

④ Information sharing

⑤ outreach & cooperation

GMOs

Types
① First Generation GMOs
→ agriculture traits, ↑ yield
eg Bt cotton, ~~golden rice~~.
② Second gen GMOs
→ focuses on nutritional quality
eg golden rice, Vit A
③
1. Agriculture. → A responsible group monitoring projects

RCGM and GEAC

RCGM (Review committee on Genetic manipulation) and GEAC (Genetic Engineering Appraisal committee) are
→ oversee the research, development & commercialization of genetically or genetically modified organisms (GMOs)

1. RCGM
→ regulatory body in India that oversees research & development in genetic engineering & stock.
→ It operates under the Dept of Biotech.

Functions:
① Approval of project & ensure safe & comply
② Risk assessment:
③ guidelines Development:
④ Monitoring:
⑤ Public Awareness:

↳ under department of Biotechnology

FSSAI (Food safety and Standards Authority of India)

Functions: Regulatory oversight
① Testing for Hazards
② Quality control
③ Adulteration check
④ Safety & consumer Protection.
⑤ Shelf-life & Sensory Evaluation
⑥ Promoting Industry standards

Types of Licences: 1. Basic Licence
< 12 lakhs
Small vendors

2. State Licence
12 < x < 20

3. Central Licence >20
(Large business, Import/export)

Compliance Rules of FSSAI:
① Regular Food Testing
② Proper Labelling
③ Sample Packaging
④ Accurate Record-Keeping
⑤ Lab Standards
⑥ Consumer Protection

Importance of FSSAI:
① Ensures food safety
② Consumer Protection
③ Standardized Food quality
④ Prevents Food Adulteration
⑤ Regulation of Food business
⑥ Promotes Transparency
⑦ Encourage International Trade
⑧ Boost Brand Reputation for Business
⑨ Prevents Food Fraud.
⑩ Enhances Nutrition health.

• HACCP (Hazard Analysis Critical Control Point)

Principles:
① Conduct a Hazard Analysis
② Determine CCPs (cooking, cooling, packing)
③ Establish critical limits (min, max) → temp, pH, time (165° chicken)
④ Establish monitoring procedures → temp check, pH
⑤ Establish corrective Actions
⑥ " Verification Procedure
⑦ " Record-Keeping & Documentation Procedure.

Importance:
① Prevents foodborne illness (food poisoning, bacteria)
② Ensures Regulatory compliance
③ Enhances Consumer Confidence
④ Reduce food safety Hazards
⑤ Global trade facilitation.

• Food Analysis & Testing in lab:
Nutrition, Safe, good quality, legal to sell
① Nutritional content Analysis [Proximate analysis (moisture), Vitamin & mineral analysis]
② Microbiological Testing: [pathogen testing, spoilage Testing] (yeast, mold)
③ chemical contaminant testing: [Pesticide Residue Testing, Heavy Metal Testing, Mycotoxin Testing]
④ Physical property Analysis:
[Texture Analysis, colour Analysis, Moisture content Testing]
⑤ Allergen Testing
[ELISA, PCR]
⑥ Sensory Evaluation
[Descriptive Sensory Analysis]
⑦ Shelf life testing
[Accelerated shelf life testing, Microbial growth analysis, chemical deterioration Testing]
⑧ Quality Assurance & Regulatory Compliance
[Standard Testing Protocols, Batch Testing]

4. Fulfillment System (chemical, biological)
5. Minimum material procurement to completion.

Unit-4

- Food Processing Techniques

methods used to convert raw food into safe, tasty, long-lasting products.

→ Increase shelf life → Improve safety
→ Enhance taste, texture & nutrition
→ Reduce wastage

main classifications:

① Physical methods
② Chemical methods

① 1. Thermal Processing
e.g Pasteurization, Sterilization, Blanching

2. Freezing and chilling
e.g freezing, chilling

3. Dehydration / Drying
e.g Air drying, Sun drying or Spray drying Freeze drying

4. Irradiation
e.g used for spices, meats, and some fruits

5. Mechanical Processing
e.g Milling, Emulsification

6. Fermentation (Biological Processing)
[Alcoholic Fermentation, Lactic Acid Fermentation]

② 1. Fermentation
e.g yogurt, alcoholic beverages,

2. Acidification
e.g pickling, citric acid, cheese

3. Salting (curing)
e.g Cured meat, salted fish, pickling

4. Smoking (chemical Reaction)
e.g Smoked meat, Smoked cheese

5. Use of Preservatives
e.g Sodium benzoate, Sodium nitrate, Sorbic acid

6. pH control agents
e.g CaOH, phosphoric acid

7. Enzymatic Browning
e.g Amylase, Protease

8. Addition of Antioxidants
e.g VitC, BHA, VitE

9. Hydrogenation [Shelf life]
e.g Margarine, Snack food

10. Emulsification [Smooth texture]
e.g Icecream, mayonnaise

11. Sugar preservation
e.g Jams, Syrups, candied fruits

- Importance:
① Increase shelf life
② Minimized food wastage
③ Seasonal availability
④ Pathogen elimination
⑤ Reduction of toxin
⑥ Safe Packaging
⑦ Retention of Nutrients
⑧ Flavour enhancement
⑨ colour Preservation.

Additional Techniques:

1. Milling and Grinding [flour milling, [coffee grinding]

2. Extrusion:

3. Homogenization: fat particles in liquid [Milk homogenization, Fruit juice]

4. Filtration & centrifugation: [Filtration & centrifuge]

5. Microwave processing.

- **Food Hygiene**

Keeping food clean & safe during prep handling, storage, cooking

Types of food Hazards:

① Physical Hazard: foreign object like glass, metal, hair, stone or insects accidentally entered

② Chemical Hazards: Harmful pesticides, chemical, heavy metals, cleaning agents, additives
eg Pesticides, Melamine, BPA, Lubricants, Toxins, mycotoxin, veterinary drugs

③ Biological Hazards:
Bacteria, viruses, Parasites, Fungi, Prions,
Salmonella, E.coli, Listeria — Norovirus, Hepatitis A, Rotavirus — Giardia, Toxoplasma — Mold, yeast

E.coli → Bloody diarrhea
Salmonella → fever + diarrhea
Listeria → Dangerous for pregnant women
Campylobacter → Stomach cramps
clostridium → Food poisoning

Prevention of Food contamination:
① Hand hygiene
② Safe storage
③ Cooking thoroughly
④ cleaning & sanitizing
⑤ safe sourcing
⑥ Temp control
⑦ Safe thawing
⑧ Educating staff
⑨ Monitoring
⑩ Labelling

- **Biochemical Analysis for food**

Studying nutrients & chemical components of food to check:
→ quality → Nutritional value
→ safety → Legal compliance.

types:
① Nutritional Analysis
[Proximate analysis, calorimetry]
② Protein Analysis
[Kjeldahl method, Biuret Test]
③ Fat Analysis
[Soxhlet Extraction, Gas chromatography]
④ Carbohydrate Analysis
[Anthrone Test, HPLC]
⑤ Vitamin Analysis
[HPLC, Spectrophotometry]
⑥ Mineral Analysis
[Atomic Absorption Spectroscopy, Inductively coupled Plasma]
⑦ pH measurement
[pH meter]
⑧ Enzyme Activity Analysis
[ELISA, Spectrophotometric Assay]
⑨ Microbial Analysis
[culture method, PCR]

The 5Ms of GMP:

1. Machine
2. Method
3. Medium
4. Material
5. Workforce

The 5Ps of GMP:

1. Products
2. Processes
3. Procedures
4. Premises
5. People

- Good Laboratory Practices (GLP)

→ Testing is accurate, reliable & properly documented
→ Documenting all lab activities
→ Ensuring proper sampling techniques.
→ Testing compliance with regulatory standards
→ Ensuring a stable testing environment.
→ Technical transfers of validated methods b/w labs.

- Good Agricultural Practices (GAP)

→ ensures food safety right from the farm level.
→ focuses on how crops are grown & harvested

Why needed?

→ due to Dirty water
→ Unsafe pesticides
→ Poor worker hygiene

∴ There should be
→ safe farm location
→ Controlled use of pesticides
→ clean water and fertilizers
→ worker hygiene during harvesting

- Good Hygiene Practices (GHP)

cleanliness & hygiene whenever food is handled.

→ food factories
→ Kitchens
→ storage areas
→ Transport vehicles

Prevention of contamination

4-star confident

- Best Aquaculture Practices (BAP)

ensures seafood is produced safely, ethically & suitably.

→ Fish farms
→ Hatcheries
→ Feed mills
→ Processing plant

Aquaculture can harm: environment, animal welfare, Food safety

BAP focuses on:

→ environmental Protection
→ Ethical labor practices
→ Fish health & welfare
→ Food safety

∴ safe & responsible seafood production.

Unit-5

- **Food additives:**

extra substance added to food
→ make it last longer
→ taste better
→ look good
→ have proper texture

e.g
chips lasting months → preservatives
Bright coloured candies → clr additives
→ They are controlled as too much can be harmful.
→ So governments set safe limits (FAO, WHO, FSSAI, FDA)

classification

flavour, appearance, texture or shelf life.

1. Preservatives: microorganisms ✗, shelf life ↑
   e.g Sodium benzoate, sorbic acid
2. Antioxidants: oxidation ✗, rancidity
   e.g Vitamin C, vitamin E
3. colorants (Food Dyes): Enhance appearance
   e.g Tartrazine (yellow), Allura Red
4. Flavour Enhancer: Boost natural Flavour
   e.g Monosodium glutamate (MSG)
5. Sweeteners: low-calorie substitute for sugar.
   e.g Aspartame, stevia.
6. Emulsifiers: mix, as oil & water
   e.g Lecithin, diglycerides
7. Thickening agents: viscosity ↑
   e.g Xanthan gum, pectin
8. Stabilizers: uniform dispersion, separation ✗
   e.g. Carrageenan, guar gum
9. Leaving Agents: Produce gases, dough rise, airy texture
   e.g Baking soda
10. Acidulants: acidity to food, enhance flavour.
    e.g citric acid, malic acid.
11. Anti-caking agents: prevent clumping in powdered foods
    e.g silicon dioxide, calcium silicate.
12. Humectants: Retain moisture in food products
    e.g glycerol

12. Gelling agents: form gels in foods, providing texture
    e.g. Gelatin, agar.

RANK
DATE / /
PAGE

Importance

1. Preservation
2. Enhancing Appearance
3. Improving Texture & consistency
4. Flavour Enhancement
5. Nutritional Benefits
6. Consumer convenience

Limits: FAO, WHO, FSSAI, FDA

1. Acceptable daily intake (ADI): max intake daily (mg/kg bw)
2. Maximum permitted level (MPL): highest conc. in food
3. No observable adverse effect level (NOAEL)

e.g Sodium benzoate: max 0.1% in soft drinks
Aspartame (Sweetener): ADI of 50 mg/kg bw
Tartrazine (colorant): ADI of 7.5 mg/kg bw
colouring agent: 100 mg/kg in candies
flavour enhancer: MSG 0.5%
Emulsifier: 1% in baked goods

- **Food Safety & Ethics**

1. Bioethics
   → moral issues related to biology, medicine, food & biotechnology
   Key aspects:
   1. Sustainability: produce food without depleting resources
   2. Animal welfare: humanely
   3. Food safety
   4. Food security: enough nutritious food
   5. Biotechnology: GMO
2. Clinical Ethics
   → healthcare settings, nutrition, and dietary practices
   Key aspects:
   1. Patient Autonomy: Patient right to choose
   2. ~~Informed consent~~
   3. Cultural sensitivity: Respect
   4. Nutritional interventions → elderly people
   5. Conflict of interest

0	1	2	3	5	7	9	11
P_1	P_2	P_3	P_2	P_4	P_5	P_1	P_5

P_1 — 4
P_2 3 7
P_3 10
P_4 Antibiotics & Hormones
P_5 1. Antibiotic Resistance: given to animal, bacteria comes in human
result: medicines stop working, longer illness,
2. Hormonal Disruption: to make animal grow faster, it affects human body also
effects: early puberty, cancer
3. Food safety concern:
→ meat contain residues of:
- Antibiotics
- hormones

if limit exceeded → health risk
4. Allergic Reaction: rashes,
5. Public Health issues: diseases from animals to human
6. Consumer awareness: prefer organic food

RANKA
DATE / /
PAGE

(3) Research ethics
→ to ensure research is honest, safe & fair
used in: food research, human studies, ~~Animal studies, environment studies~~

Key aspects:
(1) Informed Consent
(2) Confidentiality
(3) Integrity of Research (No cheating, fake data, plagiarism)
(4) Animal welfare (3Rs): Replacement, Reduction, Refinement
(5) Environment consideration
(6) Conflict of Interest

(2) Food Production & Economic

Food production = from farm to plate
includes: growing, harvesting, processing, transporting, selling

Importance:
(1) Employment opportunities
(2) Economic growth
(3) Food security
(4) Export Revenue
(5) Support to other industries
(6) Rural Development
(7) Innovation & Tech
(8) Cultural & social impact

challenges:
(1) Climate change: food shortage & price rise
(2) Resource Management: overuse of water & soil can destroy future farming
(3) Market fluctuation & Prices fall: farmers suffer if price fall
(4) Access & equity: Not everyone gets: land, money, market

(3) Food and Drug Admin (FDA)
→ U.S agency, safe, effective
→ started in 1906
→ works under (HHS)

Key functions:
(1) Food safety Regulation.
→ controls all food except: meat, poultry, some eggs
→ sets limit for pesticide residues
→ inspect factories
(2) Drug Approval & Regulation
→ approves new medicine
→ NDA checks for safety
→ Post-market surveillance: warnings side effects
(3) Medical Device Oversight
→ controls machine used in hospital
→ 3 classes:
(1) class I → low risk (bandage)
(2) class II → medium risk
(3) class III → high risk (pacemaker)
→ Approval
510(k)
PMA

- Food Preservation

① Refrigeration & freezing
② Canning
③ Drying
④ Fermentation
⑤ Salting
⑥ Pickling
⑦ Pasteurization
⑧ Irradiation
⑨ Smoking
⑩ Preservatives
⑪ Vacuum Packing

Three levels

Primary Packaging:
direct contact with food
e.g yogurt cup, cereal box liner, a can of soup

Secondary Packaging:
several units of primary packaged food; easier handling & distribution
e.g cardboard box with cans

Tertiary:
bulk packaging for storage, handling
e.g pallet loaded with cases of beverages.

- various food Packaging Methods

1. Aseptic Packaging: Food & packaging sterilized separately
e.g MILK, purées and liquid eggs
2. Modified Atmosphere packaging (MAP): Atmosphere inside altered [O_2↓, N_2↑]
e.g Packing fresh meat
3. Vacuum Packaging: Air ✗
4. Canning: heat + canned
[canned vegetables, soups & fish]
5. Shrink wrap & Stretch wrap
eg fruits, vegetables
6. Active Packaging: that interact with food to extend shelf life
e.g Packets inside snack that absorb moisture
7. Edible Packaging
eg Edible film on fresh fruits

- Novel Packaging

1. Biodegradable Packaging (PLA)
2. Edible Films & Coating
3. Nanotechnology in Packaging
4. Self-cooling & Self-Heating Pack
[self-heating coffee]
5. Smart & Intelligent Packaging
[real time info]

- Smart Packaging methods [Labels or sensors]

① Time-Temperature Indicators (TTIs)
[dairy, seafood]
② Freshness Indicators
③ QR codes & RFID Tags
④ Oxygen Scavengers.
⑤ Moisture Absorbers [silica gel]

- Good Processing Practices

Rules & Systems at each stage
farm → factory → lab → hygiene → quality

Key elements:
① Facility Design
② Process Validation
③ Documented Procedures
④ Personnel Responsibilities
⑤ Record keeping.

(4) Cosmetic Regulation
→ Safety standards

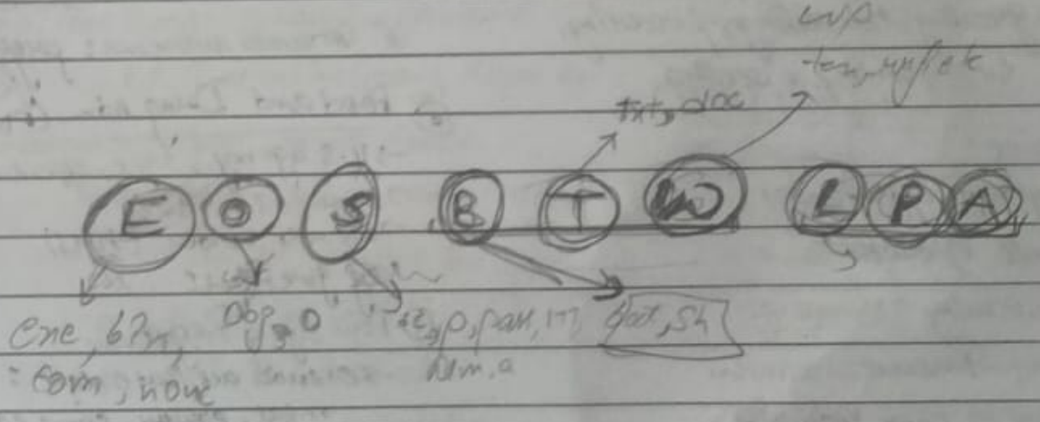
(5) Tobacco control

(6) Biological Regulation

(7) Nutrition & Labelling

Challenges

(1) Emerging Tech : hard to regulate quickly
(2) Public Perception : must balance fact vs fear
(3) Resource allocation : needs : money, staff, time
: To inspect & monitor everything



RANKA
DATE / /
PAGE

2. GEAC

→ responsible for approving the commercial release of GMOs and products derived from them.

Functions:

① Commercial Approval
② Safety Assessment → cool students forgot cold pizza
③ Policy Implementation.
④ Coordination
⑤ Public Consultation:

⑥ Institutional Biosafety Committee (IBSC)

→ vital regulatory body established in India. It is a key group that ensures safety in means.
→ protect human health. &

objectives:

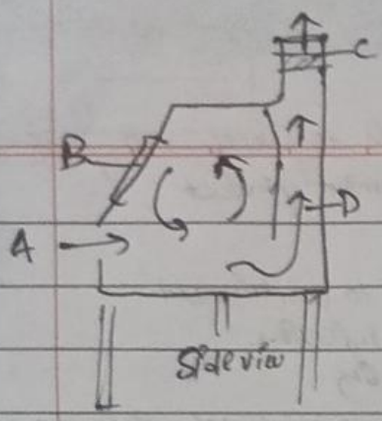
① Safety Assurance:
② Regulatory compliance:
③ Public Trust

Functions:

① Proposal Review: → Real money's How vide pops.
② Monitoring & Oversight:
③ Training & Capacity Building
④ Incident Reporting
⑤ Public Awareness

Institutional Biosafety Norms:

1. Regulatory Framework
2. Biosafety levels
3. Risk assessment
4. Good Laboratory Practices (GLP):
5. Containment measures.



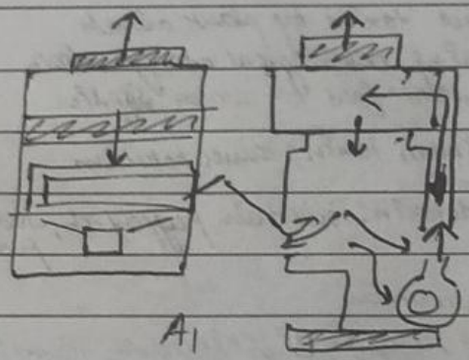
② class 2

→ Protects personnel, environment & product

→ Fan on top, creates a curtain of sterile air

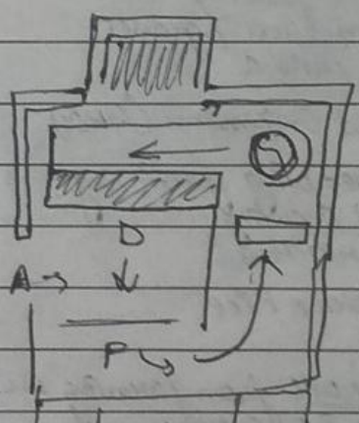
→ Air move downwards under work surface then rise up HEPA

- Type A1:
  - → Minimum inflow speed: 75 ft/min
  - → Air is mixed & can be recirculated & exhausted
  - → Not for hazardous.

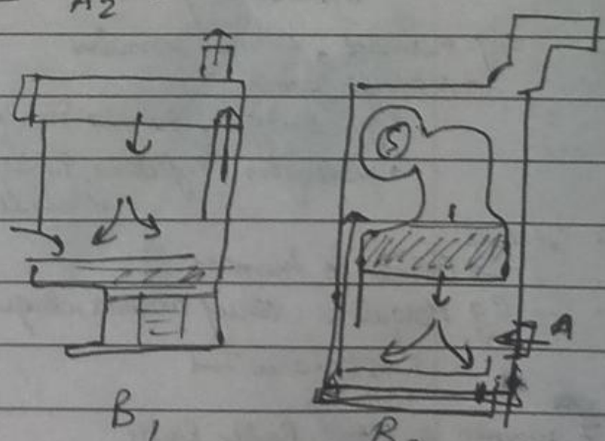


- Type A2:
  - → Minimum inflow speed: 100 ft/min
  - → 60% to 70% is recirculated, 30 to 40% exhausted.

- Type B1:
  - → uses single-pass airflow to control hazardous vapours
  - → 40% air recirculated, 60% exhausted
  - → dedicated exhaust duct.



- Type B2:
  - → 100% of air is filtered & exhaust outside.
  - → NO air is recirculated within the cabinet
  - → Ideal for tasks with toxic vapours.
  - → High risk agents (BS2–3) such as HIV, influenza.



③ class 3

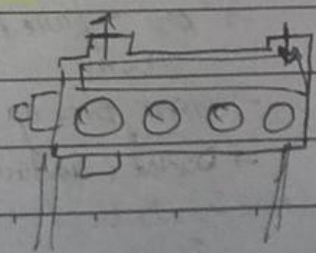
→ Protects everything

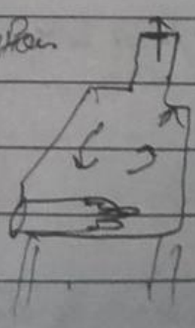
→ ~~Attached gloves~~ [Glovebox]

→ Transfer chamber (sterilize product before loading)

→ exhaust air filtered through double HEPA filters and treated with incineration

eg Ebola





④ class ④

fully body covered